

Kannan & Chatterjea Contractions

Fixed Point Theory and Applications

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An Introduction to Nonlinear Analysis and Fixed Point Theory

Lecture Notes on Nonlinear Analysis

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Chapter 2c: Kannan & Chatterjea Contractions

Generalizations of Banach's Contraction Mapping

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Motivation: Beyond Banach Contractions

Central Question

Can we relax Banach's contraction condition while still guaranteeing unique fixed points?

Specifically: **Must the mapping T be continuous?**

- Banach's contraction $d(Tx, Ty) \leq k \cdot d(x, y)$ implies T is continuous
- Question: Are there discontinuous contractive mappings that still have unique fixed points?
- Answer: YES! (Kannan 1968, Chatterjea 1972)

Definition (Key Insight)

By replacing the Banach condition with alternative contractive conditions, we can:

- 1 Eliminate the continuity requirement
- 2 Still guarantee unique fixed points in complete metric spaces
- 3 Preserve convergence of Picard iterations

Historical Context

- **1968** — Kannan proved fixed point theorem for “Kannan’s contraction”
 - Answered: Does there exist a non-continuous contraction with unique fixed point?
 - Answer: Yes, this condition works!
- **1972** — Chatterjea discovered an independent condition
 - Different from Kannan’s approach
 - Also guarantees unique fixed point
- **1977** — Rhoades observed independence
 - Banach’s (BC), Kannan’s (KC), and Chatterjea’s (CHC) are INDEPENDENT
 - None implies another
 - But all guarantee unique fixed points
- **Later** — Further generalizations by Reich, Ćirić, and others

Kannan's Contraction Definition

Theorem (Kannan - Theorem K)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ be a mapping such that there exists a number $r \in [0, \frac{1}{2})$ such that

$$d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)]$$

for all $x, y \in X$. Then T has a unique fixed point in X .

Key observation: The RHS depends on distances from x to Tx and y to Ty .

Interpretation

Instead of directly comparing distances between points, we compare distances between points and their images. This creates a fundamentally different structure.

Constraint: $r < \frac{1}{2}$ is **necessary**!

- Cannot have $r \geq 1/2$ in general
- This is the critical threshold for the theorem to hold

Example: Kannan's Non-Continuous Mapping

Example (Example 5.18 from Pathak)

Let $X = \mathbb{R}$ with usual metric $d(x, y) = |x - y|$. Define:

$$T(x) = \begin{cases} \frac{1}{3} & \text{if } x \leq 2 \\ \frac{1}{4} & \text{if } 3 < x \leq 4 \end{cases}$$

Analysis:

- T is **NOT continuous** at $x = 2$ (jump discontinuity)
- For $x, y \in [0, 3]$:

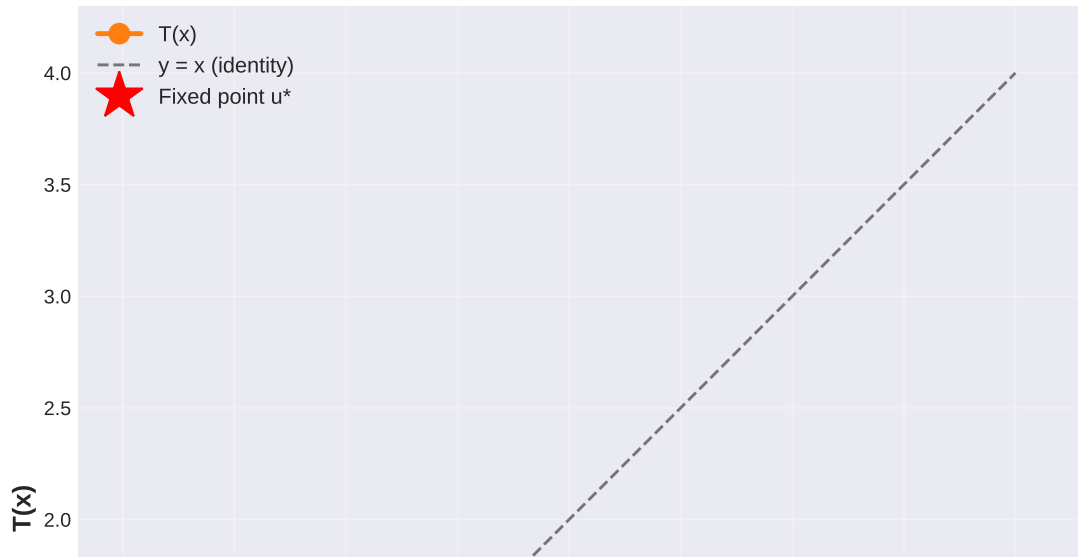
$$d(T(x), T(y)) = 0 \leq \frac{1}{3}(d(T(x), x) + d(T(y), y))$$

- Satisfies Kannan's condition with $r = \frac{1}{3}$
- Has unique fixed point $u^* = \frac{1}{3}$

Lesson: Kannan contractions do NOT require continuity!

Geometric Interpretation of Kannan's Contraction

Example 5.18: Kannan's Contraction (Discontinuous)



Chatterjea's Contraction Definition

Theorem (Chatterjea - Theorem C)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ be a mapping such that there exists a number $r \in [0, \frac{1}{2})$ such that

$$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)]$$

for all $x, y \in X$. Then T has a unique fixed point in X .

Comparison with Kannan:

Kannan (KC)	Chatterjea (CHC)
$d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)]$	$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)]$
"Same sides"	"Cross sides"

Key Difference

Chatterjea's condition uses a **cross pairing** of points and images, which creates a distinct geometric structure from Kannan's condition.

Rhoades' Independence Theorem (1977)

Theorem (Rhoades)

*Banach's contraction (BC), Kannan's contraction (KC), and Chatterjea's contraction (CHC) are **mutually independent** conditions.*

What does independence mean?

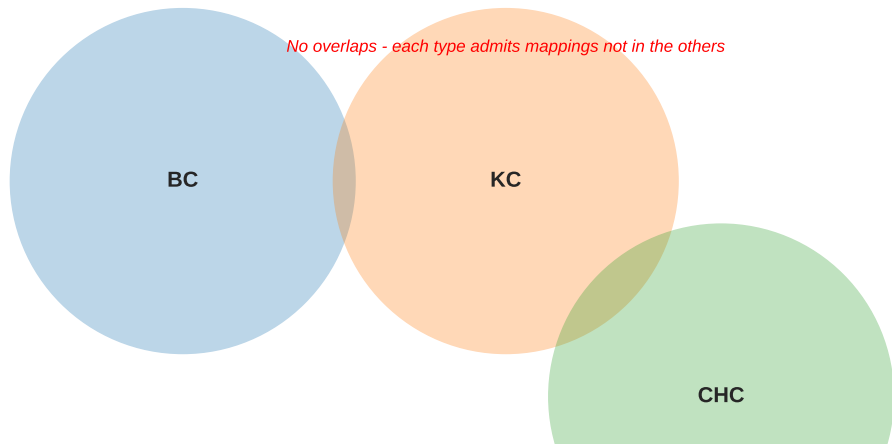
- There exist mappings satisfying BC but not KC or CHC
- There exist mappings satisfying KC but not BC or CHC
- There exist mappings satisfying CHC but not BC or KC
- No one condition implies any other

Significance

Despite being independent, all three conditions guarantee:

- 1 Unique fixed point in complete metric spaces
- 2 Convergence of Picard iterations
- 3 Solution in settings where other approaches may fail

Independence of Contraction Types (Rhoades, 1977)



Side-by-Side Comparison

Comparison of Contraction Conditions

Banach (BC)

$$d(Tx, Ty) \leq k \cdot d(x, y)$$

Contraction constant $k < 1$

T is continuous

(Uniform convergence)

Chatterjea (CHC)

Kannan (KC)

$$d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)]$$

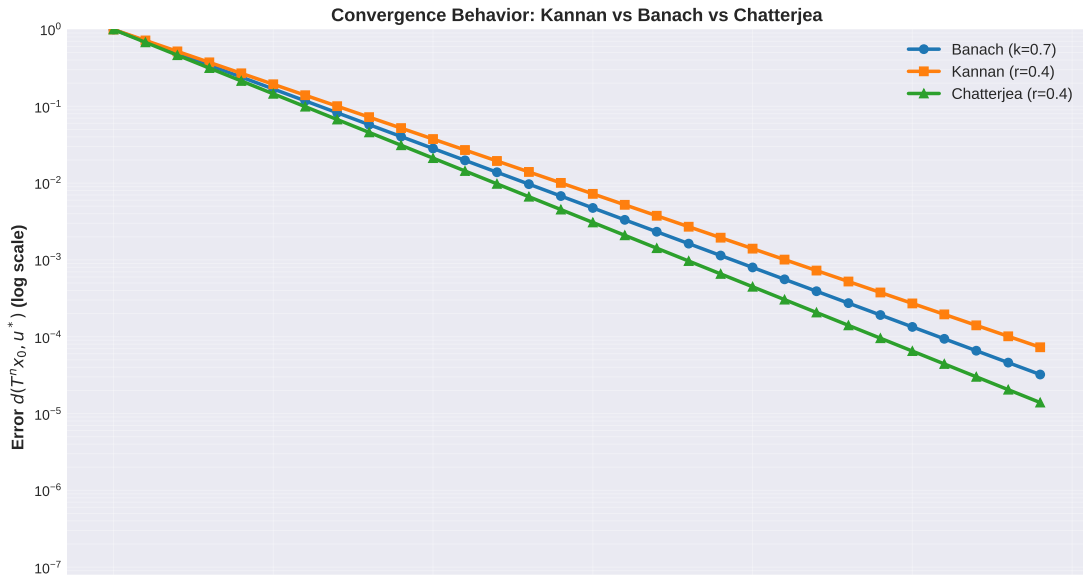
Coefficient $r \in [0, \frac{1}{2})$

T is NOT necessarily continuous

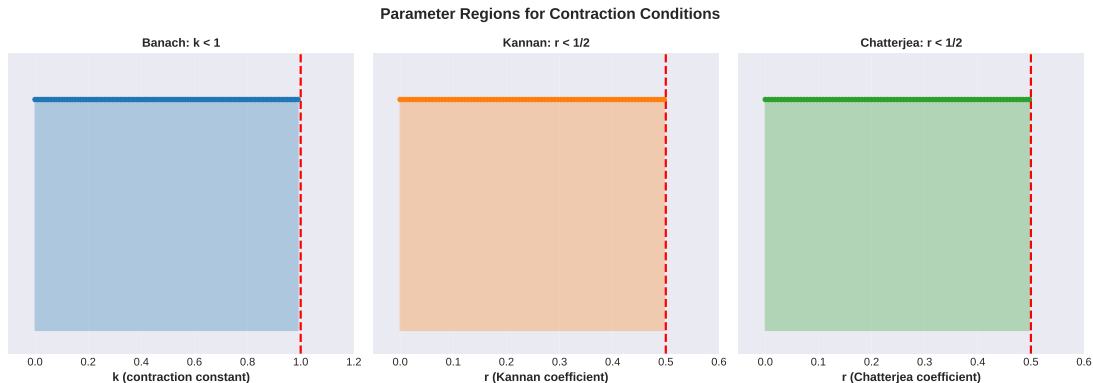
(Local convergence)

Rhoades (1977)

Convergence Behavior



Parameter Regions for Contractions



Key difference:

- Banach: $k < 1$ (more permissive)
- Kannan: $r < 1/2$ (more restrictive)
- Chatterjea: $r < 1/2$ (more restrictive, same as Kannan)

Further Generalizations of Contractions

By 1972-1977, several other contractive conditions were introduced:

❶ **Meir-Keeler (1969):**

$$\forall \varepsilon > 0, \exists \delta > 0 : \varepsilon \leq d(x, y) < \varepsilon + \delta \implies d(Tx, Ty) < \varepsilon$$

❷ **Reich (1971):**

$$d(Tx, Ty) \leq ad(x, y) + bd(Tx, x) + cd(Ty, y)$$

where $a, b, c \in [0, 1)$ and $a + b + c < 1$

❸ **Ciric (1971):**

$$d(Tx, Ty) \leq \max\{d(x, y), d(Tx, x), d(Ty, y), d(Tx, y), d(Ty, x)\}$$

❹ **Zamfirescu (1972), Hardy-Rogers (1973), and others...**

Landscape of Generalized Contractions

Generalized Contraction Mapping Conditions

**Banach
(BC)**

$$d(Tx, Ty) \leq k \cdot d(x, y)$$

**Kannan
(KC)**

$$d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)]$$

**Chatterjea
(CHC)**

$$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)]$$

**Meir-Keeler
(MK)**

ε - δ localized Banach

**Reich
(RC)**

$$d(Tx, Ty) \leq ad(x, y) + bd(Tx, x) + cd(Ty, y)$$

**Ciric
(CRC)**

$$d(Tx, Ty) \leq a_1 d(Tx, x) + a_2 d(Ty, y) + a_3 d(x, y) + \dots$$

Contraction Constant and Rate of Convergence

Definition (Contraction Ratio)

For Kannan's contraction with coefficient r , the mapping satisfies:

$$d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)] \quad (r < \frac{1}{2})$$

Theorem (Convergence Rate)

If T is a Kannan contraction with coefficient r , then:

- The Picard iterates $\{T^n x_0\}$ converge to the unique fixed point u^*
- Rate of convergence: **exponential**
- Estimate: $d(T^n x_0, u^*) \leq \lambda^n d(x_0, u^*)$ for some $\lambda < 1$

Key fact: The rate λ depends on r and the geometry of X .

Numerical Example: Kannan's Contraction in $\mathbb{R}$

Example (Implementation)

Consider $T : [0, 2] \rightarrow [0, 2]$ defined as:

$$T(x) = \frac{x}{2} + \frac{1}{4} \cos(x)$$

This satisfies Kannan's condition with $r = 0.4$ on $[0, 2]$.

```
# Compute fixed point using Picard iteration
import numpy as np

def kannan_T(x):
    return 0.5*x + 0.25*np.cos(x)

# Picard iteration
x = 1.0
for n in range(20):
    x_new = kannan_T(x)
    error = abs(x_new - x)
    print(f"Iteration {n}: x = {x:.8f}, error = {error:.2e}")
    if error < 1e-10:
        break
```

Verification of Kannan Condition

```
# Verify Kannan's contraction condition
import numpy as np
from scipy.optimize import fminbound

def kannan_T(x):
    return 0.5*x + 0.25*np.cos(x)

# Sample points
X = np.linspace(0, 2, 100)

# Verify: d(Tx, Ty) <= r[d(Tx, x) + d(Ty, y)]
r_test = 0.0
for x in X:
    for y in X:
        Tx, Ty = kannan_T(x), kannan_T(y)
        lhs = abs(Tx - Ty)
        rhs = 0.4 * (abs(Tx - x) + abs(Ty - y))
        if lhs > rhs:
            r_test = max(r_test, lhs / (abs(Tx - x) +
                                         abs(Ty - y) + 1e-10))

print(f"Required r >= {r_test:.4f}")
print(f"Condition satisfied with r = 0.4: {r_test <= 0.4}")
```

Output: Confirms Kannan condition is satisfied.

When to Use Kannan vs Chatterjea vs Banach

Property	Banach	Kannan	Chatterjea
Requires continuity	Yes	No	No
Simpler to verify	Yes	Moderate	Moderate
Parameter range	$k < 1$	$r < 1/2$	$r < 1/2$
Convergence rate	Fast	Moderate	Moderate
Common use	Standard	Special	Special
Independent from others	No	Yes	Yes

Decision Rule

- **First choice:** Try Banach's condition (simpler, faster)
- **If fails:** Try Kannan's or Chatterjea's (weaker assumptions)
- **Both KC and CHC fail?** Consider other generalizations (Reich, Ciric, etc.)

Fixed Point Theorems for Contractive Mappings

Theorem (Contractive vs Expansive)

A mapping T is contractive if

$$d(Tx, Ty) < d(x, y) \quad \text{for all } x \neq y$$

If T is contractive on a compact subset of a complete metric space, then T has a unique fixed point.

Definition (ε -Contractive)

A mapping T is ε -contractive if

$$0 < d(x, y) < \varepsilon \implies d(Tx, Ty) < d(x, y)$$

Importance

These definitions bridge Banach, Kannan, and Chatterjea contractions to even more general frameworks in nonlinear analysis.

Example (Integral Equation with Kannan Contraction)

Consider the nonlinear integral equation:

$$u(t) = \int_0^t f(s, u(s)) ds + g(t)$$

Can often be written as $u = T(u)$ where the operator T satisfies Kannan's condition (but not Banach's) under suitable assumptions on f and g .

Advantage: Kannan's approach allows:

- Relaxed continuity assumptions
- Different Lipschitz conditions
- Analysis of discontinuous kernels

Proof Strategy for Kannan's Theorem

Key Lemma

If T satisfies Kannan's condition with $r < 1/2$, then for any $x_n = T^n x_0$:

$$d(x_{n+1}, x_n) = d(Tx_n, Tx_{n-1}) \leq r[d(Tx_n, x_n) + d(Tx_{n-1}, x_{n-1})]$$

- This creates a **recursive bound** on successive approximations
- The coefficient $r < 1/2$ ensures the series $\sum d(x_{n+1}, x_n)$ converges
- Therefore, $\{x_n\}$ is a Cauchy sequence
- By completeness, $\{x_n\} \rightarrow u^*$ for some $u^* \in X$
- Uniqueness follows from the contractive property

Critical point: The constraint $r < 1/2$ is essential for this argument!

Proof Outline: Existence of Fixed Point

Existence.

① Start with arbitrary $x_0 \in X$ and define Picard sequence: $x_{n+1} = Tx_n$

② Show $\{x_n\}$ is Cauchy:

$$d(x_{n+1}, x_n) \leq r[d(Tx_n, x_n) + d(Tx_{n-1}, x_{n-1})]$$

③ Since $r < 1/2$ and the right side involves $d(x_n, x_{n-1})$, we get telescoping bounds:

$$d(x_n, x_{n-1}) \leq \lambda d(x_{n-1}, x_{n-2})$$

for some $\lambda < 1$.

④ Completeness implies $x_n \rightarrow u^*$ for some $u^* \in X$

⑤ Continuity of T at the limit (follows from contractive property) gives $Tu^* = u^*$



Note: T need not be continuous on all of X , only at the limit point.

Proof Outline: Uniqueness of Fixed Point

Uniqueness.

Suppose u^* and v^* are both fixed points with $u^* \neq v^*$.

Then by Kannan's condition:

$$d(u^*, v^*) = d(Tu^*, Tv^*) \leq r[d(Tu^*, u^*) + d(Tv^*, v^*)]$$

Since $Tu^* = u^*$ and $Tv^* = v^*$:

$$d(u^*, v^*) \leq r[d(u^*, u^*) + d(v^*, v^*)] = r[0 + 0] = 0$$

This contradiction shows $u^* = v^*$. □

Elegance: The fixed points satisfy the contraction condition themselves!

Summary of Key Points

1 Kannan's Contraction (1968)

- Condition: $d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)]$ with $r < 1/2$
- Does NOT require T continuous
- Guarantees unique fixed point in complete metric spaces

2 Chatterjea's Contraction (1972)

- Condition: $d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)]$ with $r < 1/2$
- Independent from both Banach and Kannan conditions
- Different geometric structure (cross-pairing)

3 Rhoades' Independence Result (1977)

- BC, KC, and CHC are mutually independent
- No single condition subsumes the others
- All guarantee unique fixed points

4 Further Generalizations

- Reich, Ciric, Zamfirescu, Hardy-Rogers, and many others
- Unified framework for fixed point analysis

Open Problems and Modern Developments

- **Metric Spaces:** Extensions to b -metric spaces, metric-like spaces, and generalizations
- **Multivalued Mappings:** Do KC and CHC principles extend to set-valued operators?
- **Nonexpansive and Asymptotically Nonexpansive Mappings:** Can we relax beyond contractions?
- **Partial Metrics and Order Structures:** Fixed points in ordered metric spaces
- **Computational Aspects:** Efficient iteration schemes for Kannan and Chatterjea mappings
- **Stability:** How do small perturbations affect fixed points?

Trend: Modern research increasingly focuses on weaker conditions and abstract spaces.

References and Further Reading

- **Original Papers:**

- Kannan, R. (1968). "Some results on fixed points." Bull. Cal. Math. Soc.
- Chatterjea, S.K. (1972). "Fixed point theorems." C.R. Acad. Bulgare Sci.
- Rhoades, B.E. (1977). A comparison of various definitions. Trans. Amer. Math. Soc.

- **Textbooks:**

- Pathak, H.K. (2018). "An Introduction to Nonlinear Analysis and Fixed Point Theory"
- Kirk, W.A. & Sims, B. (2001). "Handbook of Metric Fixed Point Theory"

- **Modern Surveys:**

- Recent works on generalizations in metric spaces and beyond
- Applications to differential and integral equations

Why Study Kannan and Chatterjea Contractions?

- 1 They fundamentally challenge the assumption that fixed point theorems require continuity
- 2 They provide independent tools that apply when Banach's condition fails
- 3 They demonstrate the richness of fixed point theory beyond the classical framework
- 4 They have found important applications in nonlinear analysis and differential equations

Thank you!

Questions or discussions on Kannan and Chatterjea contractions?